

Private land conservation decision-making: An integrative social science model

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ABSTRACT

Owners and managers of private lands make decisions that have implications well beyond the boundaries of their land, influencing species conservation, water quality, wildfire risk, and other environmental outcomes with important societal and ecological consequences. Understanding how these decisions are made is key for informing interventions to support better outcomes. However, explanations of the drivers of decision making are often siloed in social science disciplines that differ in focus, theory, methodology, and terminology, hindering holistic understanding. To address these challenges, we propose a conceptual model of private land conservation decision-making that integrates theoretical perspectives from three dominant disciplines: economics, sociology, and psychology. The model highlights how heterogeneity in behavior across decision-makers is driven by interactions between the decision context, attributes of potential conservation behaviors, and attributes of the decision-maker. These differences in both individual attributes and context shape decision-makers' constraints and the potential and perceived consequences of a behavior. The model also captures how perceived consequences are evaluated and weighted through a decision-making process that may range from systematic to heuristic, ultimately resulting in selection of a behavior. Outcomes of private land behaviors across the landscape feed back to alter the socio-environmental conditions that shape future decisions. The conceptual model is designed to facilitate better communication, collaboration, and integration across disciplines and points to methodological innovations that can expand understanding of private land decision-making. The model also can be used to illuminate how behavior change interventions (e.g., policies, regulations, technical assistance) could be designed to target different drivers to encourage environmentally and socially beneficial behaviors on private lands.

1. Introduction

Private land decisions influence economic, social, and ecosystem

outcomes at local to global scales, impacting the decision-maker and society at large. Here we focus on land management decisions, such as a farmer determining whether to enroll land in a conservation easement or

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plant cover crops; a woodland owner deciding whether to engage in fuel management or invasive species control; or a rancher choosing the intensity and timing of livestock grazing or whether to fence off a riparian area. These decisions have important consequences for society because they impact downstream water quality; the extent, quality, and connectivity of wildlife habitat; invasive species spread; wildfire risk; and carbon sequestration, among other societally relevant environmental consequences.

Achieving environmental sustainability relies on land and resource use patterns that contribute positively to environmental and societal outcomes. For this reason, understanding what drives human behavior (e.g., attributes of context and the individual), and how those drivers can be influenced, is critical to generating a more sustainable future (Cowling, 2014). Such understanding could improve our ability to structure incentives and other market-based tools, as well as design more effective outreach and other behavioral interventions that shift appropriate aspects of the decision context. To this end, social scientists from diverse disciplines have devoted significant effort to understand drivers of conservation behaviors on private lands (Prokopy et al., 2019; Dessart et al., 2019) and the types of interventions that encourage more sustainable land management behaviors (Byerly et al., 2018; Dessart et al., 2019; Hanley et al., 2012; de Vries and Hanley, 2016).

1.1. Diverse disciplinary perspectives

Three particularly dominant social science disciplines in the private land conservation literature with insights for individual level decision making are economics, sociology, and psychology (Bartkowski and Bartke, 2018; Pannell et al., 2006). Despite the key insights provided across these disciplines, theory on human behavior in the context of conservation decisions remains largely siloed (Cox et al., 2016). Each discipline tends to focus on different processes influencing decisions. For example, microeconomics—the study of how individuals and firms allocate scarce resources such as money and time—generally assumes that people are ‘rational utility maximizers’ who weigh the expected costs and benefits across choices, typically with perfect knowledge. These utility maximizers (previously assumed to be profit maximizers in many contexts) select the option that maximizes their utility based on their underlying preferences, which are well defined and stable over time (Levine et al., 2015; Sen, 1988). Environmental economics examines conditions under which individuals’ self-interested decisions fall short of maximizing benefits to society as a whole (e.g., market failures due to a lack of well-defined property rights, nonfunctioning markets, or mis-alignment of private and public goals), with a focus on designing incentives to correct these inefficiencies (Hanley et al., 2016).

In contrast, sociology and psychology emphasize other decision factors. Sociological analyses focus on how social interactions, norms (informal rules about how to act), social groups (such as families), and other social and institutional structures can constrain or facilitate access to information and resources and shape patterns of individual behavior and social change (Stedman, 2012). In other words, sociology considers how social structure and an individual’s position within a broader social system can shape behavior. Psychological perspectives tend to consider how an individual’s identity, emotions, values, social connections, context, and cognitive processes influence information search and interpretation, as well as the deliberativeness of the decision process. Subdisciplines of these fields can span disciplinary boundaries. For example, social psychology incorporates elements of sociological and psychological theory (Bennett et al., 2017), while judgment and decision-making and behavioral economics merge psychology and economics by explaining deviations between observed decisions and those predicted by most normative economic models (e.g., bounded rationality) (Shogren and Taylor, 2008).

The policy tools that emerge across disciplines also vary. Economics focuses on regulatory and market tools (e.g., taxation, subsidies, environmental markets) and other incentives to alter the costs and benefits

in ways that improve aggregate outcomes (Kolstad, 1999; Hanley et al., 2016). In contrast, sociology and psychology typically focus on changing social structure (e.g., shifting formal or informal institutions), social context (e.g., leveraging norms), information provision and processing (e.g., message framing), or the decision context (e.g., nudges, shifting defaults, and promoting more systematic or heuristic information processing where appropriate).

While each discipline assumes decisions involve some weighting of the perceived benefits and costs, they vary along several important dimensions. These include (1) the degree to which contextual factors are considered to constrain or enable decisions and influence outcomes; (2) the particular types of outcomes for which benefits and costs are considered; (3) whether attention is placed on the interpretation of information that can influence perception, evaluation, and weighting of outcomes; and (4) the cognitive processes by which choices are selected or decisions are made. In practice, many social scientists focus on the particular elements embraced by their discipline, while giving relatively little attention to elements outside of that domain. This tendency is beginning to shift as more scientists emerge from interdisciplinary programs broadly trained in environmental social science (Moran, 2010) and conservation social science (Bennett et al., 2017), and as behavioral economics expands into the environmental domain (Shogren and Taylor, 2008; Palm-Forster et al., 2019; Streletskaia et al., 2020). However, conceptual models of human behavior that integrate key elements of economic, sociological, and psychological theories remain scarce.

Calls for better integration across social science perspectives (e.g., Cox et al., 2016; Bruch and Feinberg, 2017; Dessart et al., 2019; Yoder et al., 2019) have identified some of the benefits of combining theories and approaches from different fields. These include theoretical and empirical innovations that generate better understanding of the processes and drivers of complex social and environmental outcomes and an improved foundation for developing public policy tools that target different drivers of human behavior (Shogren and Taylor, 2008; Dessart et al., 2019). However, the specialized training provided in most graduate social science programs and the prioritization of publishing within core disciplinary journals commonly reinforces disciplinary silos (Jacobs, 2014), discourages collaboration, and hinders understanding of the full breadth of the ‘human system’.

1.2. An integrative model of conservation decision-making on private lands

To address these challenges, we propose a conceptual model of private land conservation decision-making that integrates perspectives from three of the dominant social sciences: economics, sociology, and psychology. We developed this model as part of a collaborative process supported by the National Socio-Environmental Synthesis Center (<http://www.sesync.org/>). Our team, whose expertise spans the model’s three focal disciplines and natural science, convened in regular in-person and virtual meetings over three years during which we worked to develop a shared understanding and conceptual mapping of how our disciplines relate and contribute to understanding private lands decision-making. We acknowledge that further gains could be made by future efforts to incorporate additional social and natural science disciplines (such as those reviewed in Bennett et al., 2017), but that is beyond the scope of this effort.

Our conceptual framework highlights how the structure of the decision context, characteristics of the decision-maker, and attributes of the decision combine to shape heterogeneous conservation decisions across private lands. More specifically, the natural and built, economic, political, and social structures define a decision-making context that interacts with the individual decision-maker’s attributes, experiences, and values to shape the potential consequences associated with specific choices. Decision-makers form perceptions of those consequences and evaluate and weight them through systematic or heuristic processes to

make decisions about whether to alter their behavior. Our proposed framework recognizes the important role of feedbacks influencing how decisions and conditions evolve over time; private land conservation decisions alter environmental conditions and the micro and macro contexts to sustain behaviors or reshape future decisions.

Our framework complements recent publications that have promoted integrated models of private land conservation decisions. Both Schlüter et al. (2017) and Dessart et al. (2019) focus on individual decision-making, with the former emphasizing prominent decision frameworks, including rational actor, bounded rationality, and the theory of planned behavior, and the latter highlighting how dispositional, social, and cognitive factors affect behavior. The larger structural context of decisions, including the enabling and constraining role of socioeconomic and biophysical structure or the ways these factors operate across a decision process, are largely unaddressed. By contrast, Bennett et al. (2018) and Kemink et al. (2021) developed socio-ecological frameworks that include many components that overlap with ours but are less focused on individual decision processes. Knowler and Bradshaw (2007), Pannell et al. (2006), and Prokopy et al. (2019) review the extensive published empirical literature to identify factors that are consistently associated with conservation on private agricultural lands, yet do not situate their summary of prior research within any particular theoretical framework for decision-making. Mozato et al. (2018) and Liu et al. (2018) use their meta-analyses of quantitative studies of best management practice (BMP) adoption to propose a formal conceptual framework that recognizes the role of both macro and micro factors, but they place less emphasis on how perceptions about a BMP's costs and benefits impact decision-making processes.

Our interdisciplinary team proposes an integrated social science model of private land conservation that we hope will achieve several goals. First, we aim to improve understanding of a suite of factors and processes that influence private land conservation decisions and resulting behaviors. Second, we aspire to advance communication and research across social science disciplines by highlighting where the focal components of each discipline overlap with one another, to enable better integration in future theoretical and empirical research. Third, we intend to provide a roadmap for natural scientists seeking to better understand how social science can be integrated into collaborative work on coupled human-natural systems. Fourth, we hope to facilitate the design of more effective public action conservation tools by highlighting the range of behavioral drivers and decision-making processes that

interventions can target and how they interact.

In the following description of our model, we draw on examples from two types of private land decisions to illustrate core concepts and dynamics. The first reflects voluntary decisions by farm operators to adopt soil, water, and nutrient management practices that are important to control agricultural nonpoint-source water pollution. The second focuses on decisions by private forest landowners to use practices promoting landscape conservation values such as protecting wildlife and imperiled species. In each of these cases, private landowners and managers are making decisions regarding adoption of practices that simultaneously influence both their own and landscape-scale economic, social, and environmental outcomes. In both cases, conservation decision-making has been extensively studied via different disciplinary perspectives, but integrative social science models are rare.

2. The conceptual model

Our model focuses on individual decision-making. The decision-maker in our model is the individual or small set of individuals making a decision (also known as a behavioral intention) regarding a conservation behavior on private lands that they own or manage. The model (Fig. 1) has six main components: (A) the context for decision-making; (B) the potential consequences of the behavior for the individual and society; (C) the belief formation and decision-making processes; (D) the individual and aggregated behaviors that result from the decision-making process, and (E) the actual outcomes that occur due to patterns of conservation decisions across the landscape.

More specifically, the model highlights how the macro- and micro-structural contexts (A1–A2), individual attributes (A3), and attributes of the behavior (A4) shape both the potential consequences associated with the behavior (B) and the decision-maker's perceptions of these consequences (C1 and C2). The decision-maker evaluates and weights perceived consequences relative to their values and objectives to make decisions regarding whether to adopt the conservation practice (C3). Notably, the decision process can range from more systematic or deliberative to more heuristic or experiential and can be shaped by the decision context (A) and perceived consequences (C2). These factors also influence whether individuals engage in any active decision-making or bypass the process altogether (e.g., due to a lack of awareness, concern, or capacity). Characteristics of the individual and decision context (A) also enable or constrain the ability of the decision-maker to translate behavioral intention into behavior, which contributes to

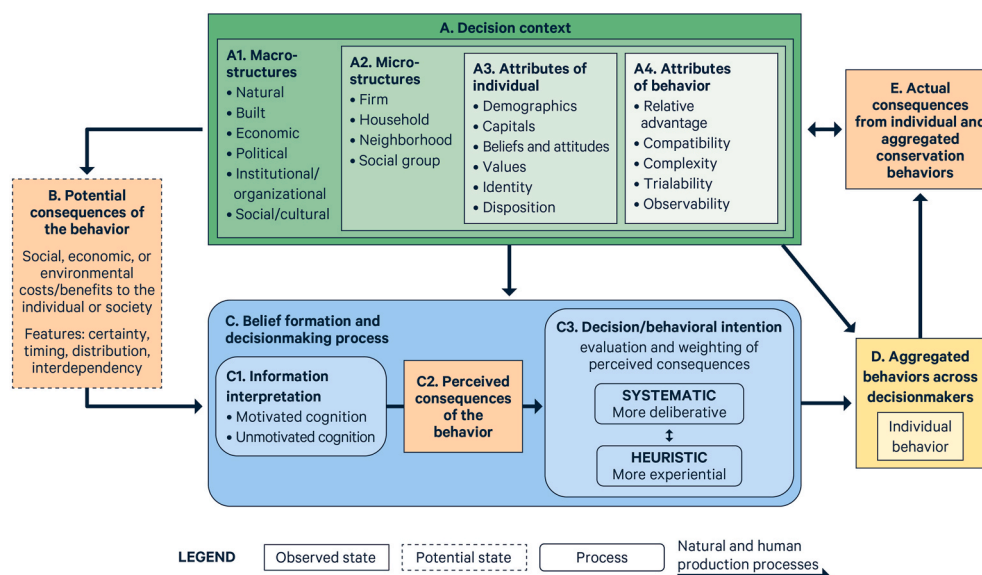


Fig. 1. Conceptual model of individual conservation decision-making on private lands.

explaining why intentions to act imperfectly predict actual changes in practice (D). The individual and aggregated behaviors based on these decisions (D) create actual consequences (E) that feed back to reshape individual attributes and the micro and macro contexts in which future decisions are made (A). Components of this framework operate at varying spatial and temporal scales, which are not explicitly represented in this model. Each component is detailed in the following subsections, emphasizing perspectives from each focal discipline (Table 1).

2.1. Potential, perceived, and actual consequences of behaviors

We begin by noting that different conservation behaviors, and the effort involved in making a decision, are assumed to generate particular material and socio-psychological consequences, or costs and benefits, that a decision-maker and other stakeholders may care about (Fig. 1, B). These costs and benefits may be financial, environmental, or social, among others. Financial consequences may include direct expenditures and returns, opportunity costs (e.g., forgone revenues), changes in land or other resource values, and tax liabilities (Epanchin-Niell and Boyd, 2020). Environmental consequences could include outcomes that the conservation behavior is designed to improve, such as water or habitat quality, and a wide array of ecosystem services, including aesthetic, recreational, production, or human health outcomes (Boyd et al., 2015). Social consequences of a behavior can include reputational effects, comfort in terms of alignment with social norms or personal values, impacts on social ties with neighbors, positive feelings from prosocial actions, and concern about the well-being of others (Bénabou and Tirole, 2003). For example, we know that agricultural land decisions have a wide range of environmental consequences for society, including direct impacts to downstream water quality due to nutrient loading (Michalak et al., 2013) and related impacts to property values, human health, and recreation (Adams et al., 2018; Backer et al., 2015; Pang et al., 2017). However, the decision-maker also may encounter personal financial and social consequences. While the potential financial consequences of taking action may be obvious, there is evidence that those engaged in conservation feel a sense of moral obligation, reflecting concern about the personal social consequences of inaction (Johansson et al., 2013; Pradhananga and Davenport, 2019; Rezaei-Moghaddam et al., 2020).

The consequences associated with a conservation behavior can vary in timing (i.e., when costs and benefits are incurred), distribution (i.e., where and to whom costs and benefits occur), and certainty (i.e., the likelihood they will occur). For example, a practice may require up-front costs, whereas its environmental or economic benefits may be delayed or accrue over time (Boyd et al., 2015; Rosa-Schleich et al., 2019), such as for cover crops in production agriculture where the soil health benefits can take 5–10 years to realize (Schipanski et al., 2014). Regarding distribution, some potential consequences may be experienced by the decision-maker, while others may have greater direct relevance for other stakeholders (Jack et al., 2008). For example, the costs of adopting forest conservation practices are often borne locally by some of the poorest communities in developing contexts, despite the benefits accruing at global scales (Poudyal et al., 2018). In terms of certainty, the potential consequences may be uncertain due to stochastic uncertainty, system complexity that prevents identifying specific outcome probabilities, and deep uncertainty about underlying cause-effect mechanisms that lead to an outcome (Dewulf et al., 2005; Morgan et al., 1990). These uncertainties can be prevalent in both environmental and human processes (Michaels and Tyre, 2012). For example, uncertainty around the consequences of individual efforts to mitigate wildfire risk or boost wildlife populations arises from the complexity in biophysical mechanisms and the dependency of outcomes on conservation practice adoption by numerous decision-makers across a landscape (Langpap and Wu, 2017; Busby et al., 2012; Epanchin-Niell et al., 2017).

We refer to the set of likely consequences if a decision-maker were to adopt a specific behavior—including the timing, distribution, and certainty of those consequences—as *potential* outcomes (Fig. 1, B). These

Table 1

Relative disciplinary emphases on various model components.

Conceptual Model Component	Scale	Disciplinary Emphasis ^a
Decision-Making Context (Box A)		
● Macrostructure (A1); microstructure (A2)	Societal/ institutional; community/ household; firm	<u>E,S,P</u> S,E: structure enables and constrains behavior E: structure shapes potential consequences S: emphasis on social and institutional structures, but considers all structures E: emphasis on economic, policy, and natural structures P: often attentive to role of social structure on information interpretation and decision processes
● Individual attributes (A3)	Individual	<u>E,S,P</u> E,S,P: shared view that attributes shape how outcomes are evaluated and weighted P: attributes affect information interpretation through motivated cognition E: attributes include preferences and resource and time constraints S: attributes position decision-makers within socially structured context
● Behavior attributes (A4)	Individual, societal	<u>E,S,P</u> S: emphasis on attributes affecting personal adoption and social diffusion processes E: emphasis on attributes shaping costs and benefits P: emphasis on attributes shaping cognition and decision-processing
Potential Consequences (Box B)		
	Individual, societal	<u>E,S</u> E: focus on objective measures of potential outcomes relevant to decision-maker's utility function and the attributes of outcomes (e.g., timing, distribution, certainty) S: focus on 'fit' between practice and attributes of decision-maker and micro-contexts
Belief Formation & Decision-Making Process (Box C)		
● Information interpretation (C1)	Individual	<u>P</u> P: perceptions are shaped by motivated and unmotivated cognition; perceptions are proximate input to decision-making; shaped by individual attributes and social connections; shaped by communication, experience, and cognitive biases
● Decision-making process (C3)	Individual	<u>E,S,P</u> E: rational actor model; expected utility theory; systematic process that accounts for resource and biophysical constraints S: social learning heuristics; social norms; rational actor P: diverse decision heuristics; spans heuristic to deliberative and is influenced by context and individual attributes
Resulting Behaviors (Box D)		
	Individual, societal	<u>E,S,P</u> E: behaviors are the output of a

(continued on next page)

Table 1 (continued)

Conceptual Model Component	Scale	Disciplinary Emphasis ^a
Behavioral Outcomes/ Consequences (Box E)	Individual, local, regional, societal	decision S: behaviors more likely when institutional and social context facilitates implementation P: behavioral intentions may or may not translate into behavior due to unanticipated constraints
		<u>E,S,P</u> E: emphasis on biophysical and economic consequences and feedback S: emphasis on social consequences and feedback P: emphasis on cognitive feedback

^a E = economics, S = sociology, P = psychology.

consequences are theoretically objective but are realized only after a decision is made and implemented. Importantly, they are rarely directly or perfectly known to the decision-maker. Instead, awareness of consequences is typically filtered by cognitive processes (C1) that shape how information is interpreted by the decision-maker. We refer to this filtered understanding of consequences as *perceived* consequences (C2), which we view as the proximate input to the decision process (C3). Finally, we recognize that there are *actual* consequences (E) associated with an individual's behaviors and with the aggregated outcomes of many individuals' behaviors (D). These actual consequences are realized after decisions are made and behaviors adopted, and they feed back to reshape the context (A) and potential outcomes (B) associated with future decisions. Potential, perceived, and actual consequences are each shaped by the decision context, individual attributes, and the specific practice, as discussed next.

2.2. Decision context

The decision context (Fig. 1, A) highlights the many factors that shape the potential and actual consequences associated with a behavior (B, E), the processes of belief formation and decision-making (C) and the translation of intentions to behavior (D). Specifically, private land conservation decisions are typically made by individuals who hold different assets, experiences, dispositions, and values (A3). These individual attributes condition the potential consequences of any decision for that specific decision-maker. In addition, decisions, behaviors, and outcomes take place within a broader macro- and microstructural context or setting (A1–A2) that can vary over time and space. Finally, the behavior in question has specific attributes that also influence the consequences (A4). We describe each of these nested levels of context in turn (also referred to as “background factors” in the reasoned action approach (Fishbein and Ajzen, 2010)), referencing examples of how context influences the potential and actual consequences (B, E), the belief formation and decision-making processes (C), as well as the potential translation of intentions to behavior (D).

At the largest, or macro, scale (A1), we recognize that private land decision-makers live in nations, states, and communities, and on landscapes that have distinctive attributes, which enable or constrain their ability to act and shape the relative costs and benefits that flow from a behavior (Hale et al., 2015; Markolf et al., 2018; Quinn and Wood, 2017). Examples of macro-level structures include natural and built, economic, political, institutional and organizational, and social structures. These combine to create locally and regionally specific conditions (or ‘structural contexts’) that shape individual decisions by altering potential costs and benefits of a behavior and how information about those consequences is interpreted.

Natural and built structures include the biophysical features of a decision-maker's land and surrounding landscape that might enable or constrain a particular behavior or influence the potential consequences associated with it (Feder and Umali, 1993; Jack et al., 2008). Examples include local weather or climate, soil quality, topography, water quality or quantity, wildlife populations and habitat quality, land use mix and land cover patterns, and the presence of different types of built infrastructure. Differences in the natural and built environment mean that adopting the same conservation behavior will provide different levels of ecosystem benefits or implementation costs (B) depending on the suitability of the location for the practice or the influences of the surrounding mosaic of land uses and conservation practices (Johnston and Bauer, 2020; Jack et al., 2008; Fleming et al., 2018). For example, the impacts of tillage or crop rotational choices on soil carbon sequestration rates are mediated by soil characteristics and climate conditions (West and Marland, 2002; Stockmann et al. 2013). Similarly, the hunting benefits provided by wetland creation or habitat restoration are influenced by environmental conditions across the broader landscape that influence wildlife populations, as well as built structure such as site accessibility (Boyer and Polasky, 2004; Knoche and Lupi, 2007). Ecological and regulatory consequences of forest owners' conservation decisions are affected by proximity of their land to nearby imperiled species populations (Mehmood and Zhang, 2005; Zhang and Flick, 2001). As another example, the presence of built water storage and distribution infrastructure shapes the ability of individuals to adopt more efficient water management practices (Hale et al., 2015).

Economic structure refers to attributes of the local and wider economy, as well as the local availability and cost of key inputs (e.g., land, labor, machinery, support services) and demand for goods produced on private lands (Jack et al., 2008). For example, commodity and input prices influence the financial outcomes of land use decisions. High crop prices can lead to disenrollment from conservation payment programs (Wu, 2000), and a high cost of planting and a high rate of return on alternative investments may reduce reforestation by forest landowners, thereby affecting provisioning of at-risk species' habitat (Zhang and Flick, 2001). Availability of supply chains that reward the use of environmentally beneficial practices can also incentivize conservation practices (Thorlakson et al., 2018; Garrett et al., 2021).

Political structure includes combined policies, programs, and governance arrangements that incentivize, facilitate, or penalize certain types of behaviors (Greiner and Gregg, 2011). A great deal of public policy is designed to stimulate conservation on private lands, through mandates, prohibitions, financial incentives, or other types of assistance to alter the costs or benefits associated with implementing a conservation practice (Hanley et al., 2012). These policies vary widely by location and include land use ordinances, environmental regulations, conservation incentive programs, and tax policies, along with public investment in programs to provide information and technical assistance regarding conservation practices (Pannell, 2008; Jack et al., 2008). The specifications of these programs and policies also can shape the availability, certainty, and timing of consequences (Hanley et al., 2012).

Institutional/organizational structures include the number and attributes of governmental and nongovernmental agencies and organizations working on conservation issues, including technical assistance providers and land-grant university extension programs (Bennett et al., 2018). Areas with more numerous, well-resourced, and better-coordinated institutions can reduce the learning, implementation, and management costs associated with adopting a new conservation practice (Prokopy et al., 2008; Pannell, 2008).

Social structures include characteristics of communities, patterns of social stratification, culture, and the normative context relative to the use of conservation (Lockwood et al., 2010; McLaughlin and Dietz, 2008). Communities with strong social capital and community attachment or low rates of outmigration among youth can reduce the costs and increase the benefits associated with conservation and other public-interested behaviors (Cho and Kang, 2017; Jones, 2010; Pretty

and Ward, 2001). Nearly all societies and communities are stratified along racial, ethnic, religious, gender, or socioeconomic lines, and these forms of social structure generate differential levels of access to benefits from conservation programs and organizations (Mountjoy, 1996). The local cultural milieu can influence the way social and environmental problems are perceived. If dominant social norms are supportive of conservation, individual decision-makers may reap social benefits from adopting conservation practices (Mastrangelo et al., 2014; Pretty and Smith, 2004). The structure of local media environments can also shape the availability and forms of information about conservation behaviors (Boykoff, 2009).

Microstructures. At the micro scale (Fig. 1, A2), our model recognizes how individual conservation decisions are also shaped by characteristics of the social or economic entities (e.g., households or firms) within which individual decision-makers are embedded. The nature and capacities of these entities help determine the potential constraints or consequences (B) associated with a conservation practice. For example, characteristics of private farms or forestry businesses (e.g., biophysical setting, scale, commodity mix, previous investments, financial assets, land ownership, and labor availability) will affect the economic costs and benefits associated with adoption of a conservation practice (Lynch and Lovell, 2003; Gedikoglu et al., 2011). The links between operation type and intentions or behavior may be obvious (e.g., practices designed to improve the environmental footprint of livestock production will be irrelevant to farms that do not raise livestock) but can also be more nuanced. One example is the impact of farm scale, as larger farms with more complex business organizational arrangements may pose greater structural environmental risks (e.g., catastrophic failures of large concentrated manure pits, higher animal to land ratios, or declines in soil health associated with the use of less diverse crop rotations) but also have greater financial and management capacity to use recommended practices effectively (Bos et al., 2013; Ren et al., 2019). Moreover, past investments in the operation, viewed as sunk costs, can create a path dependency that hinders implementing some conservation practices, such as those requiring changes to cropping systems, commodity mix, or land uses (Chavas, 1994). In addition, land tenancy, as opposed to ownership, may reduce decision-makers' engagement in conservation practices for which benefits accrue over the longer term, and non-owner agricultural operators may be more likely to make decisions that lead to soil exploitation in the short-term (Eder et al., 2021).

Another micro-scale factor reflects the household character of most private land decision-making. Households with significant off-farm or unearned income may be better able (and thus more likely) to adopt practices that do not offer short-term economic returns (Lambert et al., 2007). Conversely, the presence of children, elderly persons, or individuals with health problems requiring significant time investments may constrain an individual's ability to devote time to conservation decision-making (Som Castellano and Moroney, 2018; Hounsou et al., 2006). More generally, household decisions about land use and conservation on private lands often involve more than one decision-maker, and negotiations or coordination with other actors at the micro scale can be an important facilitator or constraint (Lyle, 2015). Relatedly, concerns about limiting the choices of children or grandchildren may prevent some landowners from entering into long term conservation easements or contracts (Kemink et al., 2021), while other studies note a potential increase in conservation on working lands when a family succession plan is in place (Duquette et al., 2012; Lubell et al. 2013).

Individual attributes. At the individual scale (Fig. 1, A3), decision-makers have certain characteristics or attributes affecting the potential consequences of their conservation behaviors (B), their belief formation and decision-making processes (C), and their ability to translate their intentions to behavior (D) (Burton, 2014; Raymond and Brown, 2011). These include both structural and cognitive attributes (Prokopy et al., 2019). Structural attributes include those conferred at birth, such as sex and race or ethnicity, and dynamic characteristics, such as knowledge, experiences, social ties, and other assets that an individual accumulates

throughout life, including financial, human, built, natural, social, political, and cultural capital (Bebbington, 1999; Emery and Flora, 2006). Cognitive attributes include a decision-maker's intrinsic instrumental and relational values (e.g., moral and environmental concern, sense of place), identity, and dispositional factors (e.g., risk tolerance, extraversion, openness to change), which generally are slow to change (Chan et al., 2016; Dessart et al., 2019). Individual attributes also include constructs such as beliefs and attitudes, which are formed and updated through the processes of information interpretation (C1).

The structural and cognitive attributes of individuals (A3) are embedded within and interact with the broader macro- and micro-structures (A1 and A2) to affect the potential (B) and perceived (C2) consequences of a decision, including an individual's ability to decide and implement a conservation behavior (C3, D) (Huang et al., 2018). For example, decision-makers who are women or members of a minority group are often less connected to local social and institutional networks in private working land settings, may have less access to information and financial support to adopt conservation practices, and otherwise must overcome barriers associated with being part of a social group having less social and political power (Gan et al., 2005; Gordon et al., 2013; Petzelka et al., 2018). Individuals with close ties to their neighbors (i.e., high social capital) may be more likely to collaborate on a conservation practice or share information about a practice (Graham et al., 2019), while individuals with greater farming experience and formal education can use their human capital to minimize the costs and increase the benefits associated with conservation (Prokopy et al., 2019). We discuss the influence of individual attributes on the perceived consequences in section 2.3.

Attributes of the behavior. Our proposed framework also recognizes that conservation behaviors differ in their attributes (Fig. 1, A4), as jointly determined by the mechanics of the specific practice and the context in which it is applied (A1–A2). These attributes of the behavior in turn contribute to shaping the behavior's potential (B) and perceived (C2) consequences. Foundational work on the adoption and diffusion of innovations highlights five influential characteristics of behaviors that predict adoption: relative advantage, compatibility, complexity, trialability, and observability (Rogers, 1995). Conservation practices that are compatible with existing farming systems, less complex, trialable or reversible, or readily observable are perceived as less risky, and often more likely to be used (Liu et al., 2018). These attributes affect the costs and benefits of implementing a behavior, including the time and effort spent in making the decision, and the level of investment and adjustments in labor, knowledge, and management required for implementation (Baerenklau and Knapp, 2007; Atwell et al., 2008).

2.3. Belief formation and decision-making process

Information interpretation. While the potential consequences of a behavior may be determined by the processes we have described, the perception of these potential consequences (Fig. 1, C2) is the proximate influence on an individual's decision to adopt or not adopt a particular practice. Perceptions or beliefs are formed through both motivated and unmotivated processes of information interpretation (C1). Motivated cognition is the unconscious tendency to process information in ways that support a particular goal, outcome, or belief (Kunda, 1990; Bénabou and Tirole, 2016). In contrast, unmotivated cognition is not biased toward a particular goal but may be biased by context and other processes (Kasperson et al., 1988; Kahneman et al., 1982). Both motivated and unmotivated cognition can result in the potential consequences being dismissed, biased, or accepted in ways that depend on the attributes of the decision-maker (A3), the structural decision context (A1, A2), and attributes of the behavior (A4), as detailed next.

When people are engaged in motivated processing, they are searching for and interpreting information in ways that support their pre-existing beliefs about what is "right," as shaped by their socially and culturally defined values or identity, previous experience or ability, or

social connections (A3) (Kunda, 1990; Bénabou and Tirole, 2016). Regarding individual attributes, someone with positive environmental values or a strong conservation identity may perceive the potential benefits of a conservation activity to be greater than the actual benefits because holding this belief helps to support their values and identity (Prokopy et al., 2019; Dayer et al., 2018). Someone with a greater tolerance for risk or higher locus of control (from a personality or dispositional standpoint) may perceive the potential costs of action as lower than they actually are, or the task of implementing the activity as easier than it is, due to their high levels of confidence (Trujillo-Barrera et al., 2016). Information may also be interpreted through cues from the decision-maker's peers or other social groups who influence the type of information that is shared and how it is presented (Kasperson et al., 1988). For example, farmers often place greater trust in their agribusiness consultants than in conservation professionals, so they may be more receptive to information from that entity and attach more importance to that information when making decisions than to information presented by an entity that does not share their values (Eanes et al., 2017).

A decision-maker's micro and macro contexts (A1 and A2) also can influence information interpretation (C1) through unmotivated processing, which can lead to systematic deviations between potential (B) and perceived (C2) consequences. Context can alter perceived consequences by making certain outcomes more salient, thereby altering expectations about magnitudes or likelihoods of outcomes. For example, the availability heuristic results in individuals perceiving greater risk or likelihood of a negative outcome if they can easily recall a relevant example (Tversky and Kahneman, 1974). We find evidence of this effect in agriculture, where farmers and agricultural advisors are more likely to believe in anthropogenic climate change and have a more positive attitude toward adaptation when they perceive variability in the weather (Arbuckle et al., 2014; Mase et al., 2015; Niles and Mueller, 2016). As another example, an individual's social context may highlight the social benefits of engaging in conservation that complies with a perceived norm (e.g., Sorice et al., 2011), while a reliance on financial mechanisms to incentivize conservation can shift a conservation practice from being perceived as a prosocial activity to a financial transaction (e.g., Banerjee and Shogren, 2012). The type, frequency, and patterns of information to which individuals are exposed are also shaped by context, including characteristics of mass and social media environments, public and private information campaigns, and the social networks within which individuals are situated (Kasperson et al., 1988). For example, in wildlife conservation, there is evidence that the media influences public perceptions and management decisions through their framing of human-wildlife conflict (Bhatia et al., 2013). In soil conservation, no-till farmers with a high level of experience and knowledge serve as key intermediaries, increasing the information flow and knowledge exchange among different clusters of farmers in a geographically distributed network (Skaalsveen et al., 2020).

Perception of consequences. Ultimately, these information interpretation processes (Fig. 1, C1), shaped by individual attributes and micro- and macrostructures (A), influence whether the information a decision-maker receives about the potential consequences holds meaning, deserves additional attention, and translates accurately into the perceived consequences of taking conservation action (including the feasibility and effectiveness of the actions at their disposal) (C2). These perceived consequences correspond to the distribution of expected realizations of state or outcome variables in an economic utility model (Eiswerth et al., 2016) and to beliefs in the theory of reasoned action (Fishbein and Ajzen, 1981) or theory of planned behavior (Ajzen, 1991), which has been updated to the reasoned action approach (Fishbein and Ajzen, 2010). Reasoned action beliefs include an individual's perception of the likely consequences of engaging in the behavior (behavioral beliefs), whether it is socially acceptable (normative beliefs), and whether the decision-maker is capable of adopting the practice (control beliefs). These perceptions are inputs into the decision process (C3).

Decision/intention formation. Generally speaking, if the perceived

benefits of a change in behavior outweigh the perceived costs, as broadly defined, then an individual may decide to implement that practice (Rogers, 1975). This decision process involves some form of evaluation and weighting of the outcomes, shaped by individual attributes (Fig. 1, A3) and contextual factors (A1–A2). In the reasoned action approach, for example, decision-makers' beliefs influence their positive and negative feelings toward a practice, as captured by their attitude toward a practice, perceived norms regarding the practice, and perceived behavioral control; these in turn influence whether the individual decides (i.e., forms a behavioral intention) to adopt a particular behavior (Fishbein and Ajzen, 2010). Within an economic model, the suite of perceived consequences of a behavior are evaluated and weighted according to the individual's preferences regarding those consequences, timing, distribution, and risk, and the behavior with the greatest expected net benefit or utility is selected (e.g., expected utility theory; Hanley et al., 2016). Economic models generally assume decision-makers maximize their personal utility (i.e., net benefits), subject to any contextual constraints (e.g., budget, biophysical setting, compatibility of the behavior; A).

Our model (Fig. 1) highlights how the emergence of a decision (i.e., behavioral intention) may be mediated by the decision process employed by the individual, which includes the potential not to engage in active decision-making. In some cases, context and information interpretation processes may prevent information about a behavior or its costs or benefits from being encountered or being registered at all, such that the individual just continues with status quo behaviors without conscious consideration of whether a new behavior should be considered. For example, individuals who do not identify as conservationists may not pay attention to information about environmental issues (C1) or may not participate in the social groups (A2) that would bring such issues to light. Some may even interact with groups that actively downplay the consequences of not engaging in more conservation (e.g., Sorice et al., 2011). Similarly, if information is received but not determined to be of sufficient concern, or if there are no perceived consequences associated with the status quo (e.g., no economic costs for the profit minded or no social disapproval for the socially aware), then a lack of concern is likely to result in no active decision and thus maintenance of the status quo (Hurst and Kreuter, 2021; Upadhyaya et al., 2021). This tendency is accentuated by a general resistance to change, such as a status quo bias (Samuelson and Zeckhauser, 1998) or lack of openness to new experiences (George and Zhou, 2001). Also, if a decision-maker has insufficient financial, time, information, or other needed resources due to the decision context (A), these realities can serve as barriers to accessing information or devoting resources to the decision-making process (C). In these cases, the status quo may be maintained through a lack of awareness, concern, or capacity to engage in decision-making (such as shown by the tendency for smaller farms to be less engaged in conservation generally; see Prokopy et al., 2019).

Alternatively, an individual may enter into a decision process that involves evaluation and weighting of the perceived consequences (C2). This process can range from more systematic to more heuristic, in line with dual process theories (Chaiken and Trope, 1999; Petty and Cacioppo, 1986). Systematic decision processes are often assumed to lead to better decisions, but evidence supports that heuristic processes can be quite adaptive and necessary in some contexts (Gigerenzer and Goldstein, 1996). For example, heuristic processes tend to be more common under conditions of uncertainty (A1, A2, A4) (Kahneman et al., 1982) or among individuals lacking the ability or motivation to engage in more systematic processes (A3) (Petty and Cacioppo, 1986), such as when the costs of information search are high and the perceived consequences of a decision are not extreme.

When engaging in a more deliberative, or systematic, decision process, an individual evaluates the perceived consequences of the available actions and places weights—implicitly or explicitly—on each consequence in terms of its relative importance (e.g., based on an individual's preferences, values, or objectives; A3). As an example, several decision-makers might engage in a process to evaluate different behaviors in

terms of their potential environmental, social, and economic consequences, accounting for the distribution, timing, and certainty of those outcomes. One may place greater weight on the environmental or social outcomes, while another may place greater weight on the economic costs in their calculation (see Upadhaya et al., 2021). Similarly, they may vary in their desire to learn through trial and error, as influenced by their identity as an innovator (A3), which in turn influences how they evaluate uncertain outcomes (A4, B) (Ranjan et al., 2019). Ultimately, these calculations would lead to reinforcement of the status quo if the costs of action are too high relative to the expected benefits or if the individual is otherwise constrained by the decision context.

Economic theory typically assumes that individuals hold fixed preferences (and hence fixed weighting of consequences or ranking of outcomes in decision-making), whereas psychological theory highlights the malleability of preferences (Slovic, 1995), including how different identities of a decision-maker can be activated in different contexts (A1–A3), leading to different values being prioritized at different times (Cohn and Maréchal, 2016). Thus, contexts that prime an individual's proenvironmental identity can augment the likelihood of selecting a more environmentally friendly option (Tate et al., 2014).

Even when engaged in more systematic decision-making, where the costs and benefits of a decision are carefully evaluated and compared across decision options, a decision-maker's evaluation of outcomes can diverge from those predicted under a rational actor model, particularly for decisions where consequences are uncertain or accrue over time (B) (see Streletskaia et al. (2020) and Wilson et al. (2016) for reviews of these tendencies for private land decision-making). For example, discounting, which is the human tendency to place less value on future outcomes than those occurring today (Frederick et al., 2002), can vary across individuals based on their time preferences (A3). Discounting has important consequences for decisions involving costs and benefits that accrue over time, and in some cases individuals' discount rates can run so high that future outcomes are worth practically nothing in the present (Frederick et al., 2002; Wade-Benzoni, 2002; Van Lange et al., 2013). For example, there is evidence that farmers can hold discount rates that range from 28% to 43%, and that the higher discount rates, often cited as a reason for failure to engage in conservation, are found among the late versus early adopters (Duquette et al., 2012). Another pattern affecting evaluation of outcomes is that individuals often place much greater weight on losses than on equivalent gains (Tversky and Kahneman, 1991). This loss aversion can mean that farmers will be more motivated to adopt a conservation behavior when avoiding losses is at stake (e.g., avoiding an extra \$30 per acre in input costs) rather than pursuing a gain (e.g., achieving an extra \$30 per acre in yield) (Cinner, 2018). Similarly, forest owners may be more willing to support efforts presented as restoring "lost" forest health as opposed to improving forest health (Wilson et al., 2012). In these ways, framing and design of policies (e.g., economic incentives) (A1) can also influence evaluation of consequences within the decision process.

When engaging in a more heuristic, or experiential, decision process, an individual may evaluate only one set of the potential consequences relevant to a particular behavior, essentially ignoring other relevant information (Simon, 1972). These approaches can be very efficient and adaptive, as they are based on past experience and learning and are generally applied under conditions of uncertainty, complexity, limited information, and time constraints (Gigerenzer, 2008). However, they can lead to systematic biases (Kahneman et al., 1982). While the potential decision heuristics are numerous, they generally range from recognition or recall-based rules (e.g., the option that is familiar is better) to one-reason approaches (e.g., the first option to meet the threshold for a consequence of interest is preferred) to those building on social intelligence (e.g., the option that others are doing is best) (Gigerenzer and Gaissmaier, 2011). For example, "imitating the successful" is a heuristic in which decisions are made by learning or copying from respected others (Bandura and McClelland, 1977). This heuristic can be effective and is often applied when payoffs are uncertain or

difficult to trial individually (A4, B) or information is difficult to access (A1–A3). There is evidence that new technologies are spread among decision-makers through such a heuristic process, where adoption occurs slowly as individuals preferentially imitate others who have particular success with the new technology (Rogers, 1995). Alternatively, if information about the success of others is not available, individuals may disproportionately imitate the behavior of the majority or rely on other forms of social learning (Fielding et al., 2008).

Individual behavior. Ultimately, the behavioral intention or decision that emerges from the belief formation and decision-making process will translate into a behavior (Fig. 1, D) when other aspects of the context (A) support the ability of the individual to act on that positive intention (Fishbein and Ajzen, 2010). Whether the decision emerges from a deliberative process that accurately considers all potential consequences or from a simplifying heuristic based on one's own experience or the experiences of others, the intention to act translates into behavior about 50% of the time (Sheeran and Webb, 2016). A common explanation for this intention-behavior gap is the actual behavioral control an individual has over taking action. For example, an individual's "good intentions" to change behavior may be stymied if their perception of their ability to engage in the behavior was inaccurate (Fishbein and Ajzen, 2010), such as due to time, resource, or other constraints. As an illustration, a decision maker might decide that the negative consequences of maintaining the status quo are too great, leading to a decision to change their behavior. However, their intention to act may not be fulfilled when they realize that they do not have eligible land (e.g., a macrostructure constraint) or lack ability to meet government program requirements given the scale of their operation (e.g., a microstructure constraint) (Prokopy et al., 2019). Overall, heterogeneity in behavior across decision-makers can be driven by differences in both their individual attributes and their context, which in turn shape the potential and perceived consequences, the decision, and the individual's ability to act on that decision and adopt a particular practice (Fig. 1).

2.4. Social and biophysical feedbacks

Individual conservation behaviors aggregate across the landscape and across time to generate actual outcomes (Fig. 1, E) that feed back into the decision context, updating conditions (A) under which future decisions are made. For example, past behaviors can alter biophysical conditions (e.g., improved water quality, greater diversity of wildlife, reduced fire risk), social and economic conditions (e.g., norm shifts, increased social learning, greater financial prosperity), individual attributes (e.g., human capital, experiences, financial capital, beliefs), and even attributes of the behavior, as methods are refined and innovation occurs (Jack et al., 2008; Bennett et al., 2017). Further, behavioral inertia can foster continuation of an existing behavior, such as through an attachment to the status quo (Polites and Karahanna, 2012), or behaviors may become habits if they are simple and repeated (Dayer et al., 2018). These feedbacks are critical, as they can shift the context and potential consequences of future decisions. For example, aggregate acreage enrollment in land retirement programs can induce market feedbacks that lead to other lands entering production due to higher commodity prices, offsetting program benefits (Wu, 2000). Alternatively, shifts in social norms can influence reputational benefits or social learning outcomes.

Temporal feedback also creates path dependencies, such that initial decisions by landowners not to engage in conservation may lead to structures that act as substantial constraints later. For example, extensive drainage of private lands has increased both agricultural productivity and the amount of nutrients lost into local waterways (Strock et al., 2010). As the negative impacts of nutrients in waterways have aggregated over space and time, retaining water on agricultural lands has become important for achieving societal conservation goals. However, adoption of some practices (e.g., filter strips, wetland creation) may be less likely because previous land drainage increased the

opportunity cost of land conversion out of agriculture. Changes in aggregate macro-scale biophysical conditions can also alter the consequences of a practice. For example, habitat restoration or reintroduction of an imperiled species by a subset of decision-makers can alter the probability of an imperiled species occurring on another decision-maker's property, influencing the costs and benefits of conservation activities or the likelihood of regulation (Epanchin-Niell and Boyd, 2020; Zhang and Mehmood, 2002). Feedback can also include governance or policy changes, adaptive management, and spatial interactions (Wilson et al., 2016; Epanchin-Niell et al., 2017, 2018). The feedbacks in our model, including social, biophysical, economic, institutional, and cognitive feedbacks, occur across varying spatial and temporal scales. These scales are not explicit in our model, as they vary across attributes and context, but they are critical for decision-making and environmental change over time.

3. Utility of this integrative social science model

Bringing together often-siloed disciplinary perspectives in one integrated conceptual model can promote more robust understanding of conservation decision-making by improving communication and research across disciplines and between researchers and practitioners. Two additional benefits of the conceptual model may be particularly relevant to conservation scientists and practitioners who are working to enable private land conservation policy and action. First, the model can guide innovative data collection and analysis approaches that capture more elements of this complex system and identify *what* factors and processes are important for understanding conservation decisions, and *why* they are important in some contexts but not others. Second, it can enable more effective targeting and design of policy to encourage decisions that improve socio-environmental outcomes by influencing the most critical factors for promoting conservation behavior. Our conceptual model can help structure the complexity of the context so the relevant aspects can be considered. While ongoing work is developing examples of applying the conceptual model to drive analysis and policy, the next sections describe three uses of the model.

3.1. Improved social science understanding of decision-making

Our proposed conceptual model can facilitate a more comprehensive understanding of private land conservation behavior and encourage researchers to consider factors and processes from outside their discipline that might otherwise be neglected (Table 1). Each discipline tends to focus on particular components of decision-making as primary objects of inquiry, while treating others as statistical controls or simply ignoring them altogether. For example, an economic analysis might focus on the financial and environmental costs and benefits to the decision-maker as influenced by contextual factors. A sociologist might explore how broader structural forces and social ties or factors like gender and race constrain access to information or to conservation programs. A psychologist might focus on how variation in identity influences beliefs about the effectiveness of a practice and subsequent behavioral intentions. Due to the complexity of interactions and the varying scales at which different factors and processes unfold, it is not realistic to expect any single analysis to account for every aspect of the decision process. However, considering how the focal explanatory variable's influence may depend on interactions with other contextual and cognitive factors can improve interpretation of results and the generalizability of empirical models. For example, many studies focusing on attitudinal components of conservation decisions fail to account for broader micro- and macro-contextual factors (e.g., biophysical conditions, local institutional capacity, economic factors, and local norms) that can vary dramatically from place to place (Mozzato et al., 2018). Similarly, many economic and sociological studies include measurement of individual and firm-level characteristics and aspects of macro context that are associated with conservation behavior but inconsistently account for the

belief formation and decision-making processes that produce the observed behaviors. More comprehensive measurement and analysis of interactions among factors and processes shaping private land conservation decisions could illuminate the mechanisms that underlie the inconsistency of findings across past studies that has been highlighted by recent reviews (see Liu et al., 2018; Prokopy et al., 2019; Ranjan et al., 2019).

3.2. Improved analytic approaches to capture complexity

Most studies are conducted using temporal and spatial scales where the macro context for decision-making lacks variation (e.g., cross-sectional surveys of decision-makers in a particular place that has relatively homogenous natural, built, political, economic, and social structure). To demonstrate how and when the macro context matters, our model would suggest that future studies should collect data from across diverse settings or observational units (e.g., watershed, agricultural district, county, census tract) using common research instruments (e.g., surveys) (Bell et al., 2019). Well-designed choice experiments (survey-based or revealed preference) can also expand the range of contexts (e.g., economic, social, or political) across which individuals' choices are observed (for examples, see Wachenheim et al., 2018; Villamayor-Tomas et al., 2019). Field experiments, in particular, offer opportunities to observe real choices in response to varying policy conditions, in contrast to something more akin to intentions observed through survey-based choice experiments (see Rosch et al., 2021 for discussion of field and lab experiment opportunities). There is also a tendency to study a particular intention or reported behavior at one point in time (for a review of common approaches, see Floress et al., 2018). Our model supports that conservation decisions should be studied as dynamic processes in contexts that evolve through time (e.g., Lawley and Yang, 2015; Zipp et al., 2017; Selinske et al., 2015). Longitudinal (i.e., panel) datasets are required to better understand the feedback mechanisms that link (1) decisions and actual consequences at one point in time to (2) changes in individual attributes and the structural context to (3) shifts in potential and perceived consequences from the same behaviors at the next point in time.

In addition to an increase in these data collection efforts, our model suggests that researchers should make greater use of analytical techniques that can test the interactions among the decision context and aspects of the belief formation and decision making process. Doing so will illuminate how the decision context influences perceived consequences and constrains or facilitates certain conservation decisions. Specifically, multilevel models (e.g., hierarchical, linear mixed effects, random coefficient regression models) contain clustered or nested data with variability at multiple levels (Garson, 2013). Examples of nestedness include individuals in neighborhoods, census tracts, states, and countries, or in the case of our conceptual model, conservation decisions of individuals living in different watersheds or management units. Latent class models of discrete choice analysis also offer an opportunity to illuminate these interactions. These models assign individuals probabilistically to classes based on their preferences regarding decision attributes or choices in-set, with preferences being heterogeneous across classes. Classification of individuals can additionally be based on individual attributes and micro to macro structures (Greene and Hensher, 2003; Train, 2009), thus enabling understanding of how these factors heterogeneously influence decisions.

Finally, structural equation modeling with latent variables (SEMLV) (e.g., causal, path, simultaneous equation models) can accommodate complex relations among multiple exogenous and endogenous variables (Bollen, 1989; Paxton et al., 2011; Wooldridge, 2016). In these approaches, researchers construct empirical models specifying hypothesized relations among variables including direct, indirect, and total effects, reciprocal causation, and feedback loops. These models include measured (manifest) and inferred (latent) variables, which can be used to measure constructs and relations between socio-psychological

concepts (e.g., values, attitudes) that directly and indirectly influence behavior (Bollen, 1989, 2012; Klöckner, 2013). For example, including individual's beliefs about the consequences of a behavior as mediators in these types of models can ensure that the contributions of microstructural factors and individual attributes are not washed out by variables that likely serve as mechanisms for their influence (Raymond et al., 2011; Pradhananga et al., 2017; and Floress et al., 2017). Of particular relevance to our conceptual model, these approaches can integrate spatial and temporal processes from different disciplinary perspectives (Denny et al., 2019; Stuart et al., 2015) and provide statistical and visual expressions of complex hypotheses that cross scales (Bollen and Pearl, 2013; Grace et al., 2010, 2015).

3.3. Implications for designing tools of public action

The mechanistic insights derived from a more integrated understanding of private land conservation decision-making are also critical for informing future changes in policy and management (Wilson et al., 2016). For example, while regulation and financial incentives are employed to better align private decisions with those producing desired societal outcomes (Kolstad, 1999; Jack et al., 2008), failures may arise from limited consideration of other factors highlighted by our model, including nonfinancial preferences, structural constraints, differences between potential and perceived consequences, and application of more heuristic (rather than deliberative) decision processes. Indeed, tools of public action can be designed to influence conservation decisions through a range of mechanisms highlighted by components in our model (Fig. 1).

One approach is to alter the decision context through policies and programs to change the potential consequences of a decision (e.g., social, economic, environmental costs and benefits) (Kolstad, 1999; Wilson et al., 2016). For example, regulations or market tools can alter economic consequences (e.g., penalties for violating permit requirements or payments via the Conservation Reserve Enhancement Program to encourage practices that improve water quality, respectively). Meanwhile, interventions focused on changing the visibility of actions (e.g., public recognition or shaming for environmental actions) are likely to create reputational or social consequences. Investments in social capital (Pretty and Ward, 2001), institutional capacity for technical assistance (Wilson et al., 2019), or efforts to engage local actors in a process of social learning (Halbrendt et al., 2014) can reduce costs by overcoming social, informational, and logistical barriers to conservation behavior.

Behaviorally based tools, on the other hand, alter how consequences are perceived or whether a more deliberative or heuristic form of decision-making is used (Fig. 1, C) (Byerly et al., 2018; van Valkengoed and Steg, 2019; Wilson et al., 2016; Reddy et al., 2017; Palm-Forster et al., 2019). For example, altering the interpretation and perception of potential consequences can be achieved through messaging that highlights consequences that are meaningful to an individual based on their particular identity (Upadhyaya et al., 2021) or by disseminating information through organizations or individuals that are trusted by focal decision-makers (Eanes et al., 2017). Leveraging heuristic decision-making can be achieved by encouraging behavior uptake by key influencers in a community, thereby promoting more widespread adoption through social connections and social learning heuristics (Rogers, 1995). Also, changing decision defaults can make the "right" choice easier, while not limiting individuals' personal agency. For example, in an online auction for conservation contracts, setting the online slider bar to start at 100% (versus 0%) cost-share contributions resulted in higher bids by decision-makers (Messer et al., 2015). Finally, promoting more systematic decision-making can be achieved through the provision of decision support tools that help decision-makers estimate the long-term consequences of different management actions (Kirilenko et al., 2007).

The appropriate set of tools to alter the potential or perceived

consequences, or the type of decision process, depends on the specific macro- or microstructural context, and the individual attributes of the decision-maker in the system (Pannell, 2008). Our model can facilitate identifying the role of these different factors. In this way, the model can improve understanding of how different tools of public action can influence decisions through multiple mechanisms, be targeted across contexts and individuals or interact with each other to increase efficacy, and even have unintended consequences through pathways that might not be considered within a single discipline's perspective.

4. Conclusion

Our conceptual model integrates perspectives from sociology, economics, and psychology about (1) the role of context in influencing the potential consequences of conservation behaviors, (2) how context influences the way information is interpreted and consequences are perceived, and (3) how perceived consequences are evaluated and weighted to make a decision. Our model also highlights the dynamic and integrated nature of decisions occurring across a landscape and how changes in behavior feed back into system changes, which in turn affect the personal and societal conditions that influence subsequent decisions. Our model is flexibly applicable across heterogeneous landscapes and decision-makers and can be used to identify more nuanced barriers to conservation practice adoption that have been neglected (for a review of behavioral economic influences that are not well studied, see Streletskaia et al., 2020). It also can be used to identify leverage points for policy tools to target, explain why interventions might not be achieving their intended goal, and suggest best practices for designing interventions and analyzing data. Greater interdisciplinary integration can facilitate evaluation of a wider range of potential barriers to private landowner conservation behaviors and identify a more comprehensive and effective suite of tools to address them.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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